

PTZrobot OTA Firmware Update Utility

User Instruction Manual (Windows PC)

Overview: FWupdate.py

This utility automates the process of updating PTZrobot firmware devices using OTA (Over-The-Air) updates.

The script will:

1. Download the latest firmware package from GitHub.
 2. Extract the OTA firmware files.
 3. Start a temporary web server on your Windows PC.
 4. Instruct the gateway to fetch and install the firmware.
 5. Display progress information during the update.
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Requirements

Before using the OTA updater, ensure the following:

Software Requirements

- Windows 10 or Windows 11
- Python 3.x installed
- Internet access

Network Requirements

- The PC and the PTZrobot gateway must be on the same network.
- The gateway must be able to reach the PC over the network.

Example: Windows PC IP: 192.168.1.50 Device Gateway IP: 192.168.1.100

First-Time Setup

The next 3 steps are only required the first time you use the updater. In the future this will all be wrapped up in a simple one step installer app which checks to see if python is already installed on the users system and if not install it before going on but for now.

Step 1: Install Python

Open Command Prompt and check whether Python is installed:

To do this type: com in the search box bottom left of your windows screen and select "command prompt?" a black command prompt window will appear.

In the window type: `python --version`

If python is already installed on your PC this will return : `Python 3.x.x`

If Python is not installed, download and install it from:

<https://www.python.org/downloads/>

During installation, ensure: ☒ Add Python to PATH is selected.

Step 2: Install Required Python Module

Open Command Prompt window:

Type: `pip install requests` or `py -m pip install requests`

This installs the HTTP library used by the update utility.

You should now have python installed and will not have to do this again.

Step 3: Create an OTA Folder

Create a folder on your computer which will be used to hold your updater and all current and future updates

Example: C:\PTZrobotOTA

Download the the updater script from here and move it into the above folder:

<https://github.com/digitalbird01/PTZrobot/blob/main/OTAbin/FWupdate.py>

Running an OTA Update

Once python has been installed this will become your starting point for all future updates.

Step 1: Open a Command Prompt

Navigate to the script folder Type: `cd C:\PTZrobotOTA` press Enter

Step 2: Start the Updater

Type: `python FWupdate.py` press Enter

Step 3: Enter the Gateway IP Address

When prompted: Enter device IP address: This should be displayed on the device OLED Example: 192.168.10.100 Press Enter.

Update Process

Once the script starts it will automatically perform the following tasks:

1. Browser Opens

Your default web browser opens automatically and begins downloading:
PTZrobot-main.zip from GitHub.

2. Firmware Package Detection

The utility watches your Downloads folder and waits for the ZIP file to arrive.

Example message:

[UPDATE] Waiting for PTZrobot-main.zip to appear in Downloads...

After detection:

[UPDATE] Found ZIP:

C:\Users\<username>\Downloads\PTZrobot-main.zip

3. Firmware Extraction

The utility auto extracts all files from the repository's: OTAbin folder.

Any old firmware files and manifests are automatically removed.

Example:

Plain Text

[UPDATE] Removed old file: gateway.bin

[UPDATE] Removed old file: manifest.json

[UPDATE] Saved manifest.json

[UPDATE] Saved gateway.bin

4. Local Firmware Server Starts on port 8000

5. Utility starts updating The device gateway checks the new manifest and decides if there is a more up to date version available. If it does not find a newer version of either the gateway or driver script for that device it will simply say this and end the update.

6. Firmware Download Progress

As the gateway downloads files from the PC, progress is displayed:

```
[SERVER] Sending file (1923476 bytes)
[SERVER] 65536/1923476 bytes (3.4%)
[SERVER] 131072/1923476 bytes (6.8%)
...
[SERVER] Transfer complete
```

Windows Firewall

The first time the script runs, Windows may display:

Windows Defender Firewall

Allow: Python to communicate on: Private Networks

(If blocked, the gateway will not be able to download the firmware.)

During the Update

Do not:

- Close the Command Prompt window.
- Stop the Python script.
- Shut down the PC.
- Disconnect the network cable/Wi-Fi connection.

The script must remain running until the firmware transfer completes.

Completion

When firmware transfer is complete:

[SERVER] Transfer complete

Allow the gateway and connected devices time to complete the firmware installation and reboot if required.

The updater can then be stopped by pressing:

Ctrl + C in the Command Prompt window.

Troubleshooting

Gateway does not update

Check:

- Correct gateway IP address entered.
- Gateway and PC are on the same network.
- Windows Firewall is allowing Python traffic.
- Firmware files were extracted successfully.